Level 4: Results

Within the Kirkpatrick model, Level 4 measures whether the behavioral changes documented at Level 3 produced measurable organizational outcomes. It is the tier that connects individual competency to business results, and therefore the tier executives care about. This document is an evaluation blueprint, not a results report: the program is a portfolio piece with no organizational deployment, so no Level 4 data exists. What follows is the complete methodology, the KPI framework, the isolation techniques, and a worked ROI example that shows how the model operates once an organization supplies its own data.

The attribution chain

Level 1 establishes that the program was perceived as relevant. Level 2 establishes that it produced knowledge gains. Level 3 establishes that the knowledge moved into workplace behavior. Level 4 observes that the organizational metrics those behaviors target then improved. The argument becomes credible only when the links connect: the program was relevant, produced understanding, changed behavior, and the outcomes those behaviors drive moved in the expected direction. Strip out Level 3 and any Level 4 result is exposed to the obvious question: how do you know the program caused it?

Why adoption metrics are not results

The dominant failure mode at this tier is the volume-ROI calculation: deploy AI tools to N employees, measure that X% activated and Y% use them weekly, multiply by a vendor-estimated time saving, and report a number with a dollar sign. It is the structural equivalent of estimating a gym's health benefit by counting badge swipes at the door, since the turnstile count says nothing about whether anyone trained. The weakness is the assumption that each unit of usage yields a uniform unit of value, which the research contradicts: savings run from about 20% to about 95% across named tasks, with most tasks between 50% and 95% (Tamkin & McCrory, 2025), and value depends on whether an interaction was augmentation or automation and how it was structured (Handa et al., 2025). The cost of ignoring this arrives as an invoice: Uber's COO reported burning the company's entire annual AI budget in four months without a clean line to user value, and within days Fortune declared the metric dead. This framework measures outcomes downstream of competent use, and treats adoption as an

input condition, not a result. The vendor research agrees. OpenAI's enterprise study concludes that rapid adoption should not be equated with immediate productivity transformation (Chatterji et al., 2026, p. 23). It also finds that the largest adopters use the tool least per employee (p. 14). The same fallacy shows at national scale. In McKinsey's 2026 survey, nearly nine in ten respondents report regular AI use and 80% report personal productivity gains. Yet 37% report any effect on their organization's earnings, a figure flat on the year despite more organizations scaling. The benefits most often cited instead are non-financial: innovation, competitive differentiation, customer and employee satisfaction (Tinkoff et al., 2026, pp. 5, 12, 14). Adoption and reported individual gain are inputs; they are not the result. The measurement problem is structural, not a failure of effort. The Atlanta Fed's December 2025 survey of US CFOs sorts reported AI impacts into two kinds. Productivity-type gains are easy to deploy and hard to measure. Cost savings and new revenue are the reverse: hard to deploy, easy to measure (Baslandze et al., as reproduced by Evans, May 2026, slide 35). Usage researchers reach the same conclusion from the supply side. The unit that matters is cost per successful outcome, not cost per token (OpenRouter and a16z, 2025, p. 54).

What is measurable at enterprise scale

A deploying organization will ask where this data comes from, and the answer cuts against the easy assumption that the AI vendor supplies it. What the major providers expose to administrators is usage telemetry, not outcomes. OpenAI's own research on that telemetry measures three things per organization-week: messages, active users, and output tokens (Chatterji et al., 2026, p. 7). Its authors state that the results do not measure work products, productivity effects, or changes in organizational routines (p. 23). ChatGPT Enterprise reports active users, message volumes, and top tools; Claude Enterprise reports conversation and message counts, projects, and seat utilization; Gemini for Workspace reports active users and per-app feature usage across Gmail and Docs. Every one of those is an adoption metric, the same volume signal this framework rejects as a results measure. The conversation content that could support a quality judgment sits behind a compliance or eDiscovery interface built for litigation, not performance evaluation.

The consequence is that the KPIs that matter do not come from the AI provider at all. Efficiency and quality measures come from the organization's own systems:

the project tracker, version history, QA and incident logs, and review workflows. Some, such as a clean count of AI-caused errors, require new instrumentation, because most organizations do not yet tag errors by cause. This is why baseline establishment is a deployment prerequisite rather than an afterthought: the provider hands the organization an adoption dashboard, and the outcomes the framework actually needs are the organization's to instrument. That instrumentation is one of the practices separating organizations that report significant AI returns from the rest. They are twice as likely to have defined processes to quantify the impact of their AI initiatives (Tinkoff et al., 2026, p. 19).

The KPI framework

KPIs fall into four categories, each tracing back to specific Level 3 behaviors. Organizations select from them based on which outcomes matter and which they already have the infrastructure to measure. The three outcome families here match what employers say they expect from training investment: enhanced productivity (77%), competitiveness (70%), and talent retention (65%) (WEF Future of Jobs Report, 2025, p. 48). Those are expectations, not measured results, which is the reason this framework measures.

Efficiency gains (read together: a faster draft that needs more rounds is not a gain)

KPI	DEFINITION	SOURCE	L3 TRACE
E1 Time-to-first-draft	Time from assignment to first draft for AI-eligible tasks	PM data (ticket/doc timestamps)	B1, B3, B4
E2 Revision cycles	Average revision rounds before acceptance	Review/approval workflow logs	B5, B6, B8
E3 Rework rate	Share of deliverables needing substantive post-delivery correction	QA / client-feedback systems	B2, B5, B8

Quality improvements (the highest-consequence category: one prevented error in a regulatory filing can outweigh a cohort's efficiency gains; inaccuracy is also the AI risk organizations most often report working to mitigate, at about 56% of high performers and 52% of others, Tinkoff et al., 2026, p. 23)

KPI	DEFINITION	SOURCE	L3 TRACE
Q1 AI-related error rate	Documented errors from unverified AI output (fabricated citations, bad statistics)	Incident reports, QA logs (needs AI-error tagging)	B2, B5, B6
Q2 Appropriate delegation rate	Share of AI-assisted tasks within the model's reliable range	Quarterly sample audits (new infrastructure)	B1, B2, B7

Adoption maturation (cultural change, the organizational analog of the Level 3 behavior change)

KPI	DEFINITION	SOURCE	L3 TRACE
A1 AI transparency rate	Share of the team openly discussing AI practices	Derived from B9 aggregated by team	B9, B7
A2 Concealment reduction	Drop from the 69% concealment baseline	Anonymous pulse survey at 6 months	B9, B12

Capability development (the strategically decisive category)

KPI	DEFINITION	SOURCE	L3 TRACE
C1 Peer coaching incidence	Documented instances of participants coaching colleagues	Derived from B11 at 90 days	B11, B9
C2 Time-to-competency for new tools	Onboarding time to a new AI tool, alumni vs. non-participants	Opportunistic, when a new tool is introduced	B1, B3, B5 (tool-agnostic)

C2 is the argument that justifies AI-literacy investment over tool-specific onboarding: if alumni onboard to each new tool faster, the program produced transferable literacy, which compounds as organizations adopt more tools through 2030. Employer plans point that way: 86% expect AI to transform their business by 2030, 77% plan to reskill existing staff to work alongside AI, and 62% plan to hire people who already can (WEF Future of Jobs Report, 2025, pp. 62–63). The spread is under way: 56% of respondents report AI use in three or more functions in 2026, up from 51% (Tinkoff et al., 2026, p. 5). The base rate is already high. Each platform shift has multiplied the tools in use, and a typical large US enterprise now runs 400 to 500 SaaS applications (Productiv, as reproduced by Evans, November 2025, slide 10).

Isolating the program's contribution

Attribution is the central methodological challenge, and the framework is honest that every technique produces an estimate, not a proof. It uses several in combination, in decreasing order of rigor: a comparison-group design where deployment is phased (the later group is a natural control); trend-line analysis against 6–12 months of pre-program baseline; and independently collected manager and participant estimates of the share of improvement attributable to the program, each adjusted downward for optimism. The phased design is a difference-in-differences comparison, the same structure Anthropic's labor-market analysis uses to separate AI effects from the business cycle: a change that appears in both groups is not attributed to the intervention, and the design's minimum detectable effect should be stated up front (Massenkoff & McCrory, 2026, pp. 11–12). The result is reported as a range, and the ROI model uses the lower bound by default. If the return is positive at the lowest defensible attribution, the case is strong; if it is positive only at the upper bound, it should be communicated as speculative.

The ROI model

The model is not invented for this project. It applies the Phillips ROI Methodology, the established framework that extends Kirkpatrick with a fifth level, financial ROI, and is the standard approach to costing training in corporate L&D (Training Industry). The formula below is Phillips' own, net program benefits over fully loaded cost expressed as a percentage; the isolation step above is Phillips' "isolate the effects of the program"; and the conservative inputs follow his Guiding Principles, which call for a fully loaded cost, the most conservative estimate among alternatives, and attribution adjusted downward for error. What this project supplies is not a new formula but defensible inputs, each drawn from the research corpus rather than a vendor benchmark.

THE ROI MODEL · TERM BY TERM

Efficiency benefit = **N** × **W** × **T_ai** × **S** × **A**

Quality benefit = **Q** × **Q_reduction** × **A**

ROI = (Total benefit − **C_program**) ÷ **C_program** × 100

N participants in the cohort (default 200)	**W** fully loaded annual cost per participant (default $140,000)
T_ai share of work hours on AI-eligible tasks (default 25%)	**S** time savings on those tasks with competent use (default 30%)
A share of improvement credited to the program (default 40%)	**C_program** total fully loaded program cost (default $300,000)
Q annual cost of AI-related quality failures (default $400,000)	**Q_reduction** reduction in those failures (default 20%)

Two independently attributed benefit streams, set against total program cost. The calculator below starts at these conservative defaults.

INTERACTIVE · ROI MODEL

ROI calculator

The defaults below are the conservative inputs from the static model, each with the rationale that earned it; together they reproduce the document's worked example. Change any input and the chain recomputes.

N Participants

200

Mid-sized cohort for a departmental rollout.

W Fully loaded annual cost

$ 140000

Mid-career base ~$100K × 1.4 fully loaded (benefits ~30% per BLS, plus payroll taxes and overhead).

T_ai AI-eligible task share

25 %

MGI puts agents at 44% of US work hours as technical potential and about 27% automated by 2030 in its midpoint scenario (Yee et al., 2025); 25% sits at the adoption end. Handa et al.'s 57% is a share of conversations, not hours, and is not used here.

S Time savings, competent use

30 %

Against an 81% in-conversation median; 30% acknowledges full-cycle verification overhead.

A Attribution factor

40 %

Lower-bound; assumes 60% of improvement is from other factors.

C_program Total program cost

$ 300000

Amortized development (~735 hrs, Chapman), delivery, ~6 hrs participant time, admin, the full four-level evaluation, and contingency; anchored to ATD 2024 benchmarks.

Q Annual AI-quality-failure cost

$ 400000

Estimated annual cost of AI-related quality failures across the cohort.

Q_reduction Failure reduction

20 %

Conservative one-fifth reduction.

Reset to conservative defaults

EFFICIENCY BENEFIT

N × W × T_ai × S × A

$840,000

QUALITY BENEFIT

Q × Q_reduction × A

$32,000

↘ ↙

Total annual benefit **$872,000**

– Program cost (C_program) $300,000

↓

RETURN ON INVESTMENT

191%

ROI = (Total benefit − Program cost) ÷ Program cost × 100

At the current attribution (A = 40%), ROI is 191%; halving A to 20% would give 45%.

Formula and conservative-input discipline follow the Phillips ROI Methodology: net program benefits over fully loaded cost, with attribution adjusted downward for error.

Timeline, and the full chain

MILESTONE	ACTIVITY
Pre-program	Establish KPI baselines (≥3 months historical) and the A2 concealment baseline
90 days	Level 3 complete; begin watching Level 4 KPIs for direction (no ROI yet)
6 months	First Level 4 point: measure KPIs, run estimation surveys, calculate a preliminary ROI range
12 months	Full evaluation: trend-line analysis, final ROI with confidence range, integrated L1–L4 narrative

Populated, the chain reads as a single argument: participants rated the program relevant and named intended changes (L1), gained measurable knowledge across all four domains (L2), changed specific behaviors that managers and self-reports both confirmed (L3), and the organizational metrics those behaviors target improved (L4). Each level hands the next its evidence; remove any one and the argument has a gap. That is why the framework was designed as an integrated system from the outset, not four instruments bolted together after the fact.[1]

[1] *No Level 4 data exists because the program has not been deployed at organizational scale. Causal certainty is unattainable outside a randomized controlled trial, which is rarely feasible organizationally; the combined isolation techniques produce a defensible estimate reported as a range, not a proof. The model requires pre-program baselines for every tracked KPI, so baseline establishment is a prerequisite for deployment, and some outcomes (cultural transparency, time-to-competency for future tools) may take longer than twelve months to fully materialize.*

Sources

- **Kirkpatrick Partners** — *The Kirkpatrick Model*
- **Phillips, J. J. / ROI Institute** — *The ROI Methodology* (the five-level model extending Kirkpatrick to financial ROI; the formula, isolation step, and conservative Guiding Principles this model follows)
- **Training Industry** — *Glossary: Phillips ROI Methodology* (the corporate L&D costing convention cited alongside the Phillips methodology)
- **Fortune** — *Uber's AI budget burn* and *Tokenmaxxing is over* (May 2026)
- **Tamkin, A. & McCrory, P.** — *Estimating AI Productivity Gains from Claude Conversations* (Nov 2025)
- **Handa, K. et al.** — *Which Economic Tasks Are Performed with AI? Evidence from Millions of Claude Conversations* (Feb 2025) (augmentation-automation split)
- **Anthropic Interviewer** — *What 1,250 Professionals Told Us About Working with AI* (Dec 2025) (69% concealment baseline)
- **World Economic Forum** — *Future of Jobs Report 2025* (Jan 2025)
- **Association for Talent Development** — *2024 State of the Industry* (per-employee and per-learning-hour L&D cost benchmarks); the worked example also uses the standard 1.25–1.4× fully-loaded employee-cost multiplier and the Chapman Alliance content-development-hours benchmark
- **OpenAI** — *ChatGPT Enterprise workspace analytics*; **Anthropic** — *Claude Enterprise usage analytics*; **Google** — *Gemini for Workspace usage reports*
- **Chatterji, A., Holtz, D., Rakholia, N., Tambe, P. & Weeratunga, G.** — *How Organizations Use AI: Evidence from ChatGPT* (OpenAI working paper, Aug 2026)
- **Tinkoff, D., Van der Veken, L., Chui, M. & Balakrishnan, T.** — *The state of AI in 2026: On the road to ROI* (McKinsey & Company, Aug 2026)
- **Yee, L., Madgavkar, A., Smit, S., Krivkovich, A., Chui, M., Ramirez, M. J. & Castresana, D.** — *Agents, robots, and us: Skill partnerships in the age of AI* (McKinsey Global Institute, Nov 2025)
- **Aubakirova, M., Atallah, A., Clark, C., Summerville, J. & Midha, A.** — *OpenRouter State of AI: An Empirical 100 Trillion Token Study* (OpenRouter & a16z, Dec 2025)
- **Evans, B.** — *AI Eats the World* [slide deck] (Nov 2025; May 2026)

- **Massenkoff, M. & McCrory, P.** — *Labor Market Impacts of AI: A New Measure and Early Evidence* (Anthropic, Mar 2026)

© Michael Ritchot. Code licensed under the MIT License. Instructional content licensed under CC BY-NC-SA 4.0.